Automation Mini Project - Ball & Beam

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Introduction

The ball-and-beam setup is an experiment whose objective is to stabilise a ball on a beam. The beam is controlled by a voltage, and the measured variables are the beam angle and the position of the ball. The position is measured using an impedance-reading circuit.

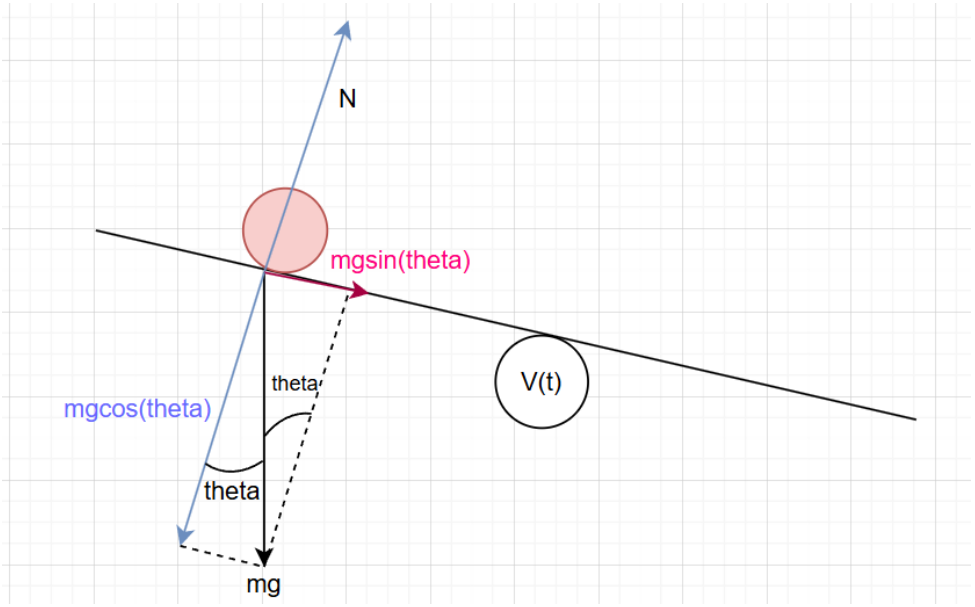


Figure 1 – Force diagram of the ball-and-beam system

Analysis of the beam system block diagram

System identification is performed in closed loop.

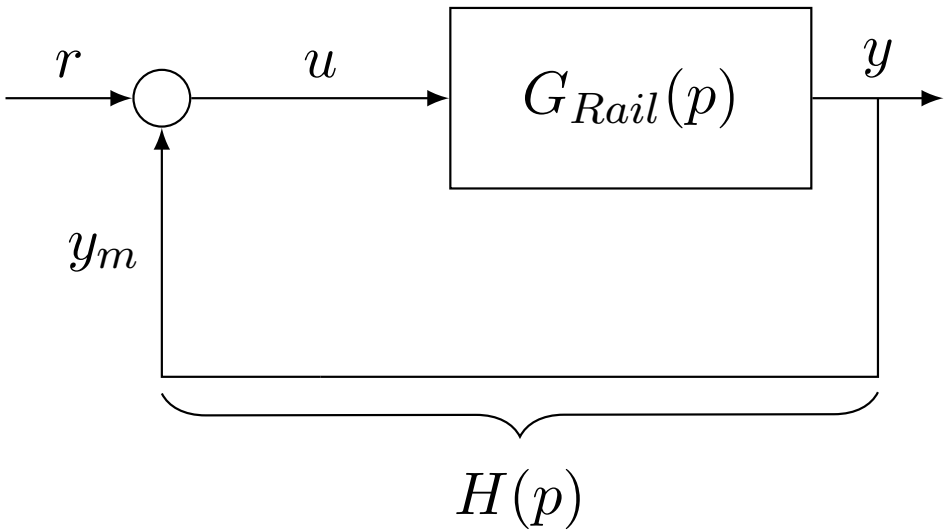


Figure 2 – Block diagram representing the beam system

Implementation of N4SID

We use the MATLAB function "multisine()" to generate a test signal for this system.

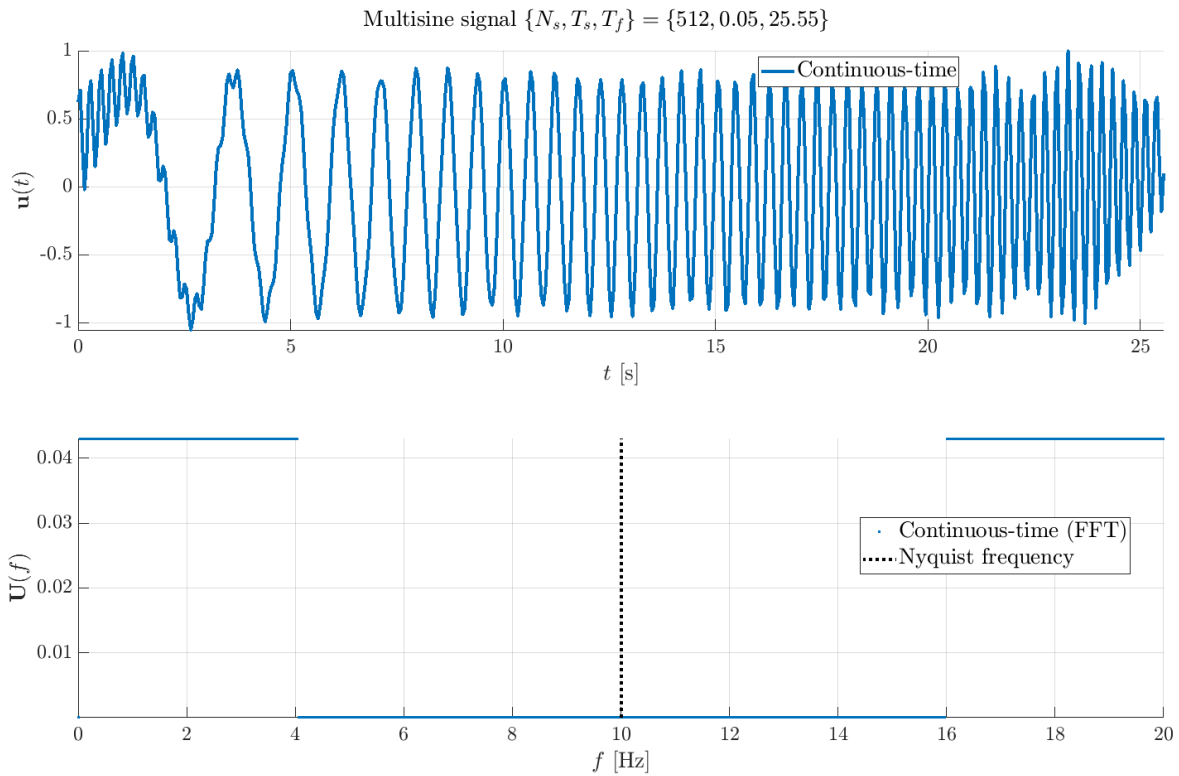


Figure 3 – The input signal, of the "multisine" type, between 0.1Hz and 4Hz

Testing the model obtained with N4SID

We obtain the transfer function using the MATLAB function "n4sid()" in both the time and frequency domains.

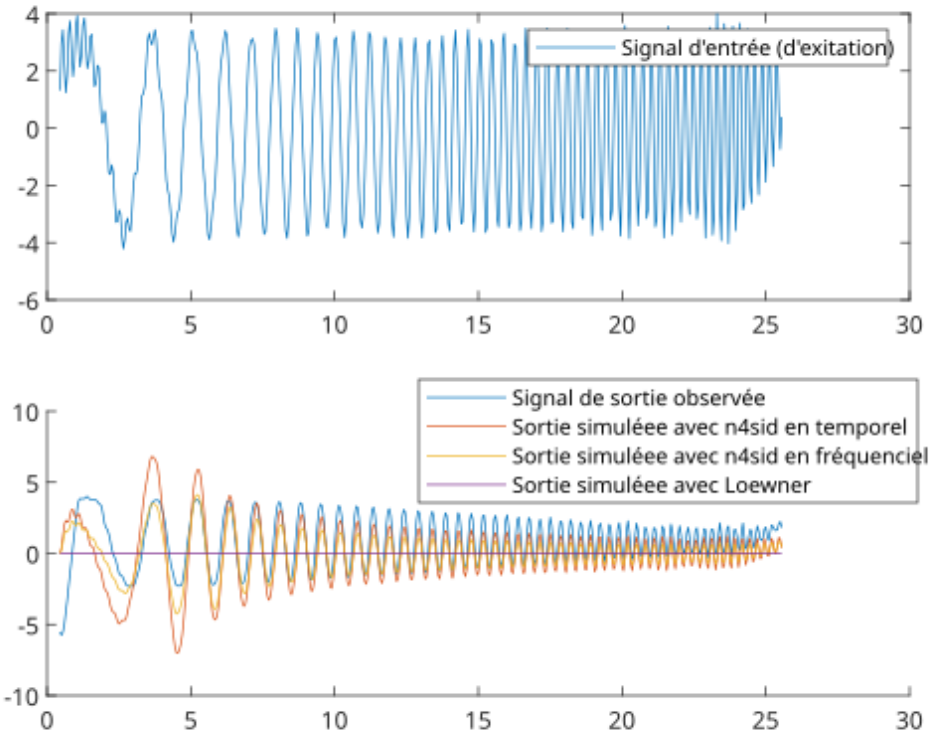


Figure 4 – Behaviour of the different models obtained

After comparison, we selected "time-domain, order 2".

Transfer function of the system : Beam

This led us to represent the beam system using the following transfer function :

$$\boxed{H(z) = \frac{0,2977z^{-1} - 0,2962z^{-2}}{1 - 1,825z^{-1} + 0,8496z^{-2}}}$$

After finding the desired model, we then obtained the actual transfer function of the beam using d2c() :

$$H(p) = \frac{G(p)}{1 + G(p)} \Rightarrow G(p) = \frac{H(p)}{1 - H(p)}$$

Calculation of the system controller : K_p

We designed PID feedback for the beam system. After discussing this with the professor, we were told that the system already includes an integrator. We therefore chose a closed-loop system with a simple P controller, as shown below :

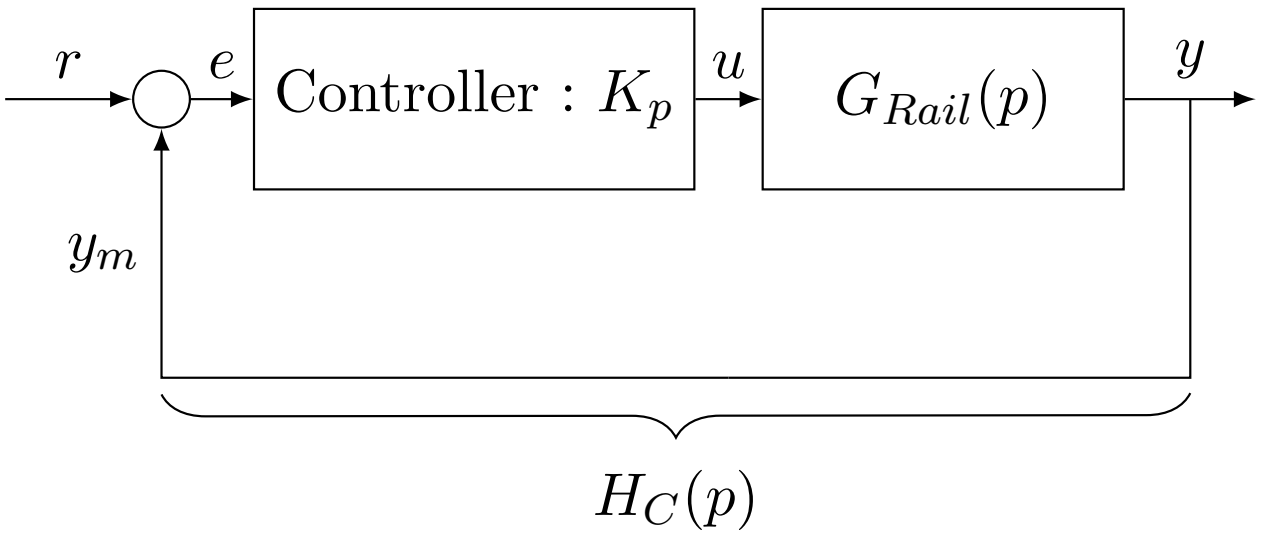


Figure 5 – Block diagram representing the beam system with controller

After identifying the system using n4sid(), we obtained the following transfer function :

$$H(p) = \frac{6,464p + 0,6686}{p^2 + 3,261p + 10,52}$$

Using the transfer function of the beam system, we recalculated the new closed-loop transfer function with the proportional gain :

$$G(p) = \frac{H(p)}{1 - H(p)} = \frac{5,564p^3 + 18,72p^2 + 60,43p + 6,058}{p^4 + 0,5397p^3 + 10,14p^2 - 0,9659p + 89,3}$$

$$G_{BF\ K_p}(p) = \frac{K_p G(p)}{1 + K_p G(p)}$$

Finally, we tested different values of P to observe the closed-loop response time with the controller.

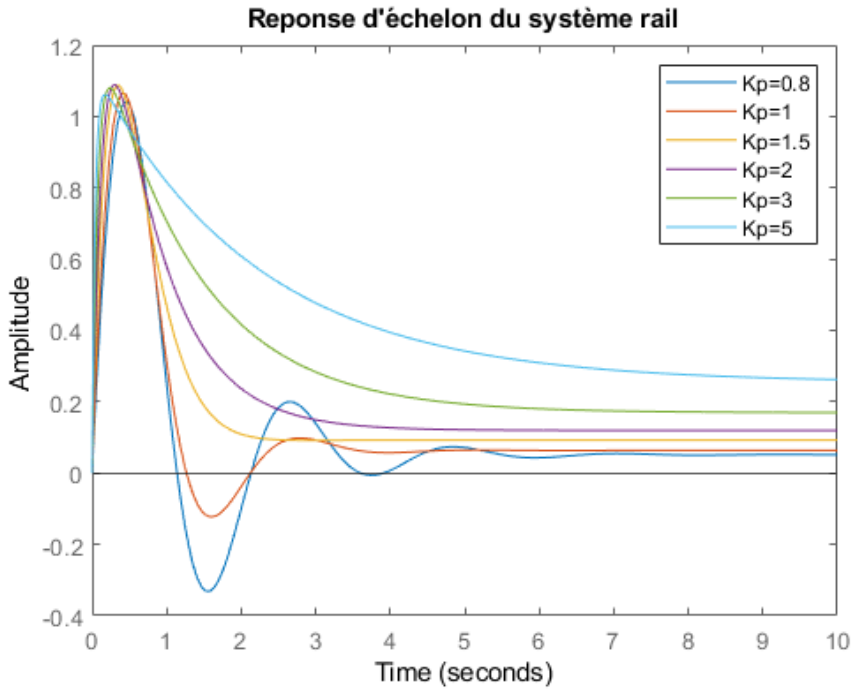


Figure 6 – Step response with different K_p values

We chose $\boxed{K_p = 1}$, which gives a fast response.

Closed-loop system with the ball

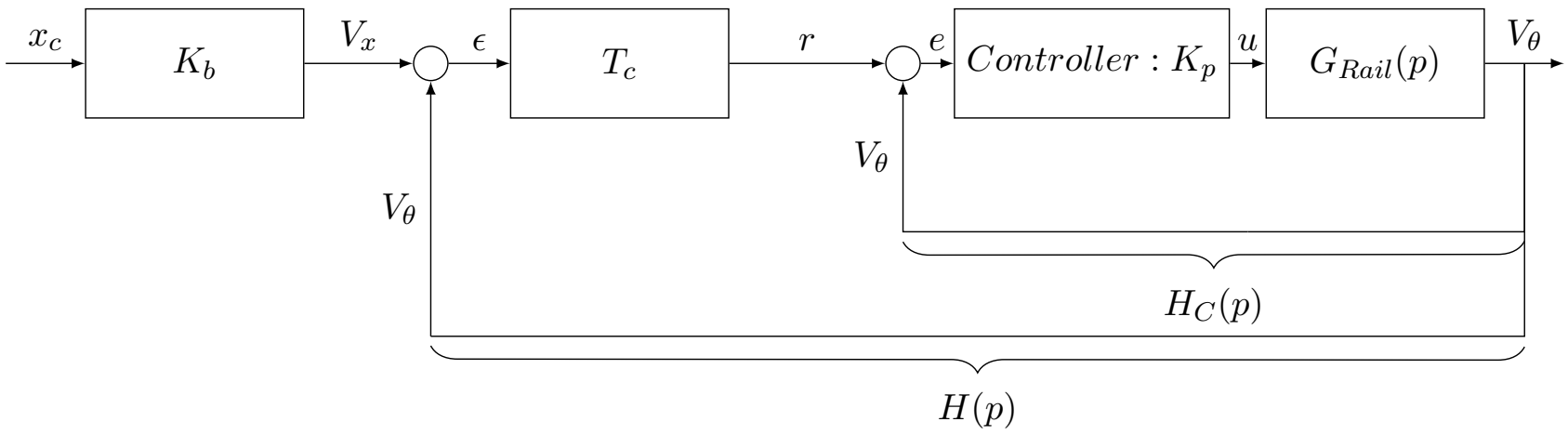


Figure 7 – Block diagram representing the complete ball-and-beam system

We would like the system input to be a desired position for the ball, between -50cm and +50cm. Let us therefore start by finding a K_b to convert the position into a voltage.

Position-to-voltage conversion

We calculated the transfer function from the mechanical equations :

$$\sum F_x = m\ddot{x} = mg\sin(\theta(t))$$

However, $\theta \approx 0$. We can therefore simplify using :

$$\sin(\theta) \approx \theta$$

This gives us :

$$m\ddot{x}(t) = mg\theta(t)$$

Then, by applying the Laplace transform, we obtain :

$$p^2X(p) = g\Theta(p) \Leftrightarrow \frac{X(p)}{\Theta(p)} = \frac{g}{p^2} \Rightarrow \boxed{\frac{V_x(p)}{V_\theta(p)} = K_b\frac{g}{p^2}}$$

To calculate K_b, we performed a linear regression.

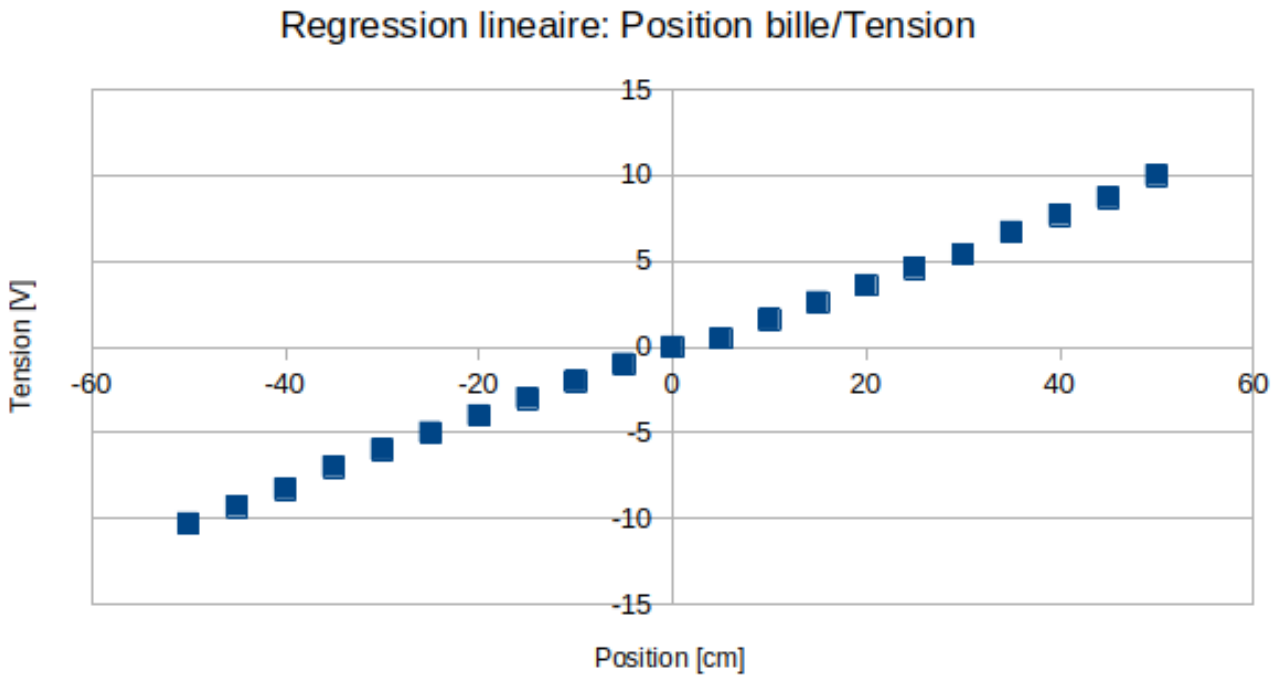


Figure 8 – Relationship between the ball position and the system output voltage for performing a linear regression

This gave us, in Microsoft Excel, $\boxed{K_b = 1.926}$.

Analysis of the important system frequencies

For this second loop of the system, we start by plotting the Bode diagram of the ball-and-beam system.

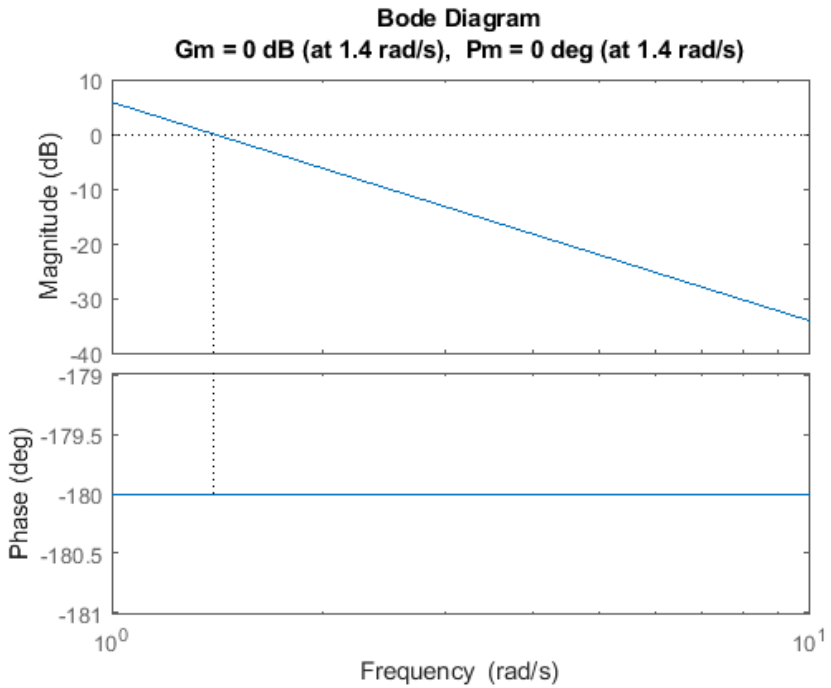


Figure 9 – Bode diagram of the beam system without the ball

The selected controller is a phase-lead compensator.

The critical point where additional phase margin is required is where the magnitude is 0dB, i.e. at 1.4 rad/s. We therefore design the phase-lead compensator around this frequency. The phase-lead compensator has the following canonical transfer function form [1] :

$$T_c(p) = K_p \frac{1 + \alpha T p}{1 + T p} \ , \ where \ \alpha \ > \ 1$$

1. https://homepages.laas.fr/fgouaisb/donnees/M1ICM/slidesM1ICMp8.pdf

Calculation of the phase-lead compensator parameters

Let us start by calculating the desired phase margin :

$$\Theta_{Margin} = \Theta_{Desired} - \Theta_{System} = 125° \Rightarrow \Theta_{Desired} = 55°$$

Next, let us calculate α :

$$\alpha = \frac{1 + \sin(\Theta)}{1 - \sin(\Theta)} = \frac{1 + \sin(55°)}{1 - \sin(55°)} \approx 10$$

Finally, let us calculate T :

$$T = \frac{1}{\omega_m * \sqrt{\alpha}} \Rightarrow T = \frac{1}{1.4 * \sqrt{10}} \approx 0,22$$

This gives us a phase-lead compensator :

$$\boxed{T_c(p) = \frac{1 + 2,2p}{1 + 0,22p}}$$

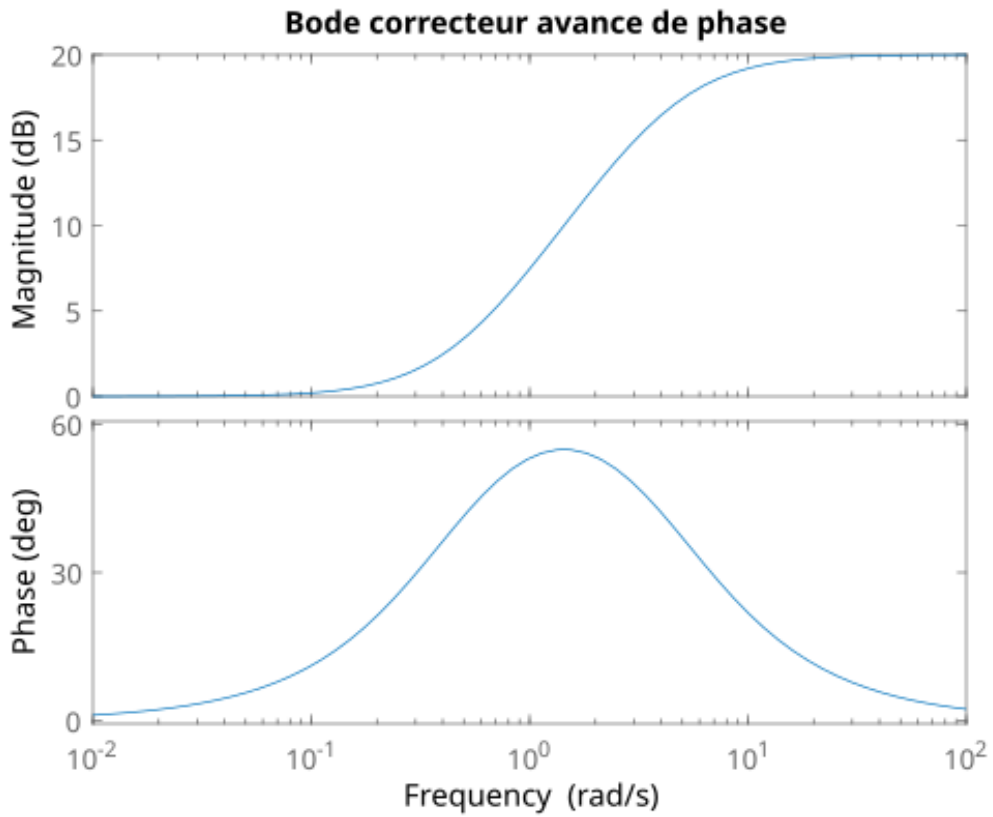


Figure 10 – Bode diagram of the phase-lead compensator

Experimental verification

→ Demonstration ←

Verification of the phase margin

Using the allmargin function, we determine the phase margin of the complete system in closed loop.

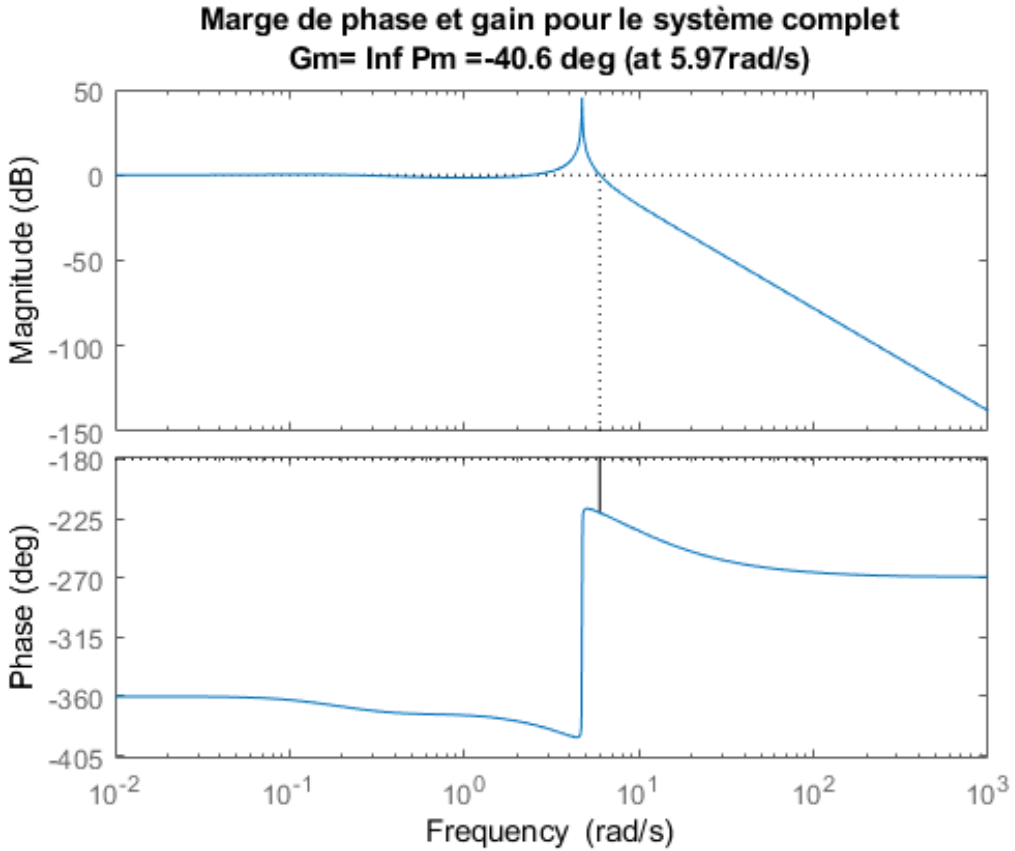


Figure 11 – Bode diagram of the complete system

Conclusion

The loop is closed and the ball is under control.

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